

Alignment-Preserving Hierarchical Hypothesis Management for Difficult Open-Ended Differential Diagnosis

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Abstract

Difficult open-ended differential diagnosis requires preserving concept identity and evidence-conditioned belief state across reasoning stages. Flat pipelines fragment synonyms, discard branch-local updates before global ranking, and conflate name binding with reasoning quality. We call this cross-stage consistency *alignment* and introduce **alignment-preserving hierarchical hypothesis management (APHHM)**, which combines a case-adaptive comparison axis, concept-level competition, evidence-conditioned score write-back, and bounded global decoding. We evaluate APHHM on fixed 100-case subsets of DiagnosisArena (Zhu et al. 2026), MedCaseReasoning (Wu et al. 2025), and Open-XDDx (Zhou et al. 2025) under an unmodified scoring interface. On DiagnosisArena, changing from a matched random axis to the case-adaptive axis within the deployed selector cascade raises Top-1 accuracy from 0.37 to 0.71. On audited open-set MedCaseReasoning endpoints, jointly disabling concept-equivalence handling lowers accuracy from 0.50 to 0.42; semantic compression removes 56.5% of candidates without reducing accuracy and exceeds a count-matched blind merge under partition randomization. On Open-XDDx, evidence-conditioned score write-back improves micro-F1 from 0.576 to 0.651 at fixed evidence budget, whereas increasing the budget changes F1 by at most 0.008. APHHM remains above a similar-call-count flat control on all three end-task metrics, while a retrospective stage audit localizes many apparent recall misses to the evaluation interface. The results show that preserving concept identity and belief state turns large hypothesis spaces into coherent decision spaces.

1 Introduction

Large language models (LLMs) can propose differential diagnoses from clinical vignettes, and retrieval-augmented generation (RAG) can enlarge the set of disease hypotheses drawn from case reports, guidelines, and textbooks (Lewis et al. 2020; Wei et al. 2022; Xiong et al. 2024). Difficult open-ended diagnosis is therefore often framed as a *recall-discrimination dilemma*: increasing coverage admits the correct disease more often, but also introduces redundant and granularity-mismatched candidates that make ranking harder.

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Coverage alone does not characterize the difficulty of a multi-stage system. Even after a correct hypothesis has been retrieved, cross-stage failures can prevent it from reaching the final output. Three cross-stage failures characterize this problem:

- **Synonym fragmentation:** near-equivalent disease strings compete as separate leaves and consume frontier budget.
- **State discontinuity:** evidence changes a branch-local score but not the belief state used by global decoding.
- **Evaluation detachment:** the ranked concept is correct, but the emitted name fails to bind to the benchmark’s canonical target.

Our central view is that open-ended diagnosis should be treated as **hierarchical hypothesis management**: candidates must be organized for comparison, compete as clinical concepts rather than strings, and share an evidence-conditioned state across local and global decisions.

Existing systems organize different components of this process. Direct chain-of-thought (CoT) and flat RAG couple retrieval, organization, and ranking in one generation step (Wei et al. 2022; Zhao et al. 2025; Xiong et al. 2025). Verification methods such as Dual-Inf improve reflection over a flat differential list (Zhou et al. 2025). Multi-agent frameworks such as MDAgents and MAC diversify computational *roles* (Kim et al. 2024; Chen et al. 2025b; Yuan et al. 2026); APHHM complements this role structure by organizing the geometry of the disease hypothesis space. Fixed specialty or knowledge-graph hierarchies provide reusable routing (Zuo et al. 2025), whereas APHHM constructs a case-specific comparison coordinate system. Interactive diagnosis benchmarks emphasize information acquisition, which is adjacent to—but distinct from—keeping a large static evidence set organized and state-consistent (Nori et al. 2025; Rose et al. 2025).

Contributions.

1. **A cross-stage formulation of open-ended diagnosis.** We distinguish concept, state, and evaluation consistency, and show why operator-sensitive endpoints and earliest-stage error attribution are necessary to evaluate them.
2. **A hierarchical method that preserves concept and state consistency.** APHHM combines a case-adaptive

comparison axis, equivalence-class-aware candidate competition, evidence-conditioned score write-back, and bounded global decoding.

3. **Mechanism-focused empirical evidence.** Comparisons across three diagnostic benchmarks support an axis-conditioned pipeline effect, concept-aware competition on audited open-set endpoints, and evidence-conditioned score write-back on Open-XDDx; APHHM retains its end-task margins under a similar-call-count flat control.

All systems are evaluated with the benchmark interfaces unchanged, and Section 4.4 separately audits name–concept binding at the evaluation interface.

2 Related Work

Direct CoT (Wei et al. 2022), self-consistency (Wang et al. 2023), self-refine (Madaan et al. 2023), Medprompt-style prompting (Nori et al. 2023), and medical RAG (Lewis et al. 2020; Xiong et al. 2024; Zhao et al. 2025; Xiong et al. 2025) improve knowledge access and list generation. Their candidate spaces remain largely flat, so recall, organization, and ranking are coupled: increasing coverage can also increase synonym crowding and mix incompatible levels of specificity.

Verification and collaboration methods address a different part of the pipeline. Dual-Inf adds reverse verification and reflection (Zhou et al. 2025); MDAgents (Kim et al. 2024), MAC (Chen et al. 2025b; Yuan et al. 2026), and MEDDxAgent (Rose et al. 2025) structure agent roles or discussion; Chain-of-Diagnosis specializes diagnostic dialogue (Chen et al. 2025a). These approaches organize computation, whereas APHHM explicitly organizes the disease hypotheses and maintains a shared state that later stages must read.

Existing medical hierarchies are typically fixed or used outside runtime reasoning. KG4Diagnosis routes through a knowledge graph (Zuo et al. 2025), DeepRare combines specialized tools for rare disease (Zhao et al. 2026), and H-DDx uses ICD structure to evaluate clinically related predictions (Lim et al. 2025). APHHM instead constructs a temporary, case-specific comparison axis and uses it during candidate generation, evidence updating, and final arbitration.

Interactive benchmarks such as SDBench optimize information acquisition (Nori et al. 2025); we instead consider static vignettes, where the evidence is already available and the central problem is how a large hypothesis set is represented and propagated. Our stage-aware evaluation separates generation, local selection, global ranking, and name binding, which standard Top- k accuracy combines into one outcome.

3 Problem Formulation

3.1 Task and Belief State

The input is a static clinical vignette V , observed facts F traceable to spans in V , and a corpus $\mathcal{K}$ shared by all retrieval-based methods. The system outputs an ordered list of at most k_{out} concrete diseases and a case-specific belief state

Earliest failed stage	Required intervention
Parent absent	Revise the case axis
Leaf absent	Improve branch-conditioned recall
Local elimination	Preserve a bounded local frontier
Global misranking	Improve state update or arbitration
Binding failure	Standardize name–concept binding

Table 1: Earliest failed stage and the intervention it implies.

$B_t = (G_t, q_t, s_t, E_t)$, containing the hierarchy G_t , concept identities q_t , relative scores s_t , and evidence annotations E_t at stage t . Every reasoning stage reads this state and returns an updated state. The objective is accurate ranking of static, text-based, single-primary vignettes under bounded retrieval and inference budgets; internal scores are relative ranking quantities rather than clinically calibrated probabilities.

3.2 Cross-Stage Consistency Requirements

Let $\mathcal{C}$ be the candidate set after recall. Increasing $|\mathcal{C}|$ can raise coverage, but the benefit depends on how candidates are represented and propagated, so we require three forms of consistency. *Concept consistency*: candidate strings denoting the same clinical concept should share one competitive slot, or else surface variants consume a budget intended for distinct hypotheses. *State consistency*: evidence used by a local stage must update the belief state read by subsequent stages, or else local reasoning cannot affect the final decision. *Evaluation consistency*: the emitted name must bind to the canonical concept expected by the scoring contract, or else a correct ranked concept can be recorded as absent. Case-adaptive organization defines the families and slots over which the first two requirements operate, and stage-aware evaluation then distinguishes failures of organization, local selection, global ranking, and name binding.

3.3 Stage-Aware Error Attribution

A failed case is assigned to the earliest applicable stage; later stages are considered only after earlier ones succeed. This ordering prevents an evaluation-interface error from being reported as insufficient recall. The cascade has five stages. *Parent absent*: no clinically acceptable Level-1 (L1) family can host the target disease. *Leaf absent*: a suitable family exists, but no equivalent Level-2 (L2) disease leaf is generated. *Local elimination*: the target leaf exists but is removed before global decoding. *Global misranking*: the target reaches the global pool but falls below the cutoff. *Binding failure*: the target ranks sufficiently high, but the scoring interface records no canonical binding. Each stage admits a different repair (Table 1), so the attributed stage, rather than the aggregate error rate, determines which component should be changed.

4 Method: Alignment-Preserving Hierarchical Hypothesis Management

Figure 1 summarizes APHHM and the state fields shared across stages. The method has three stages: case-adaptive organization, concept-level competition, and shared-state decoding.

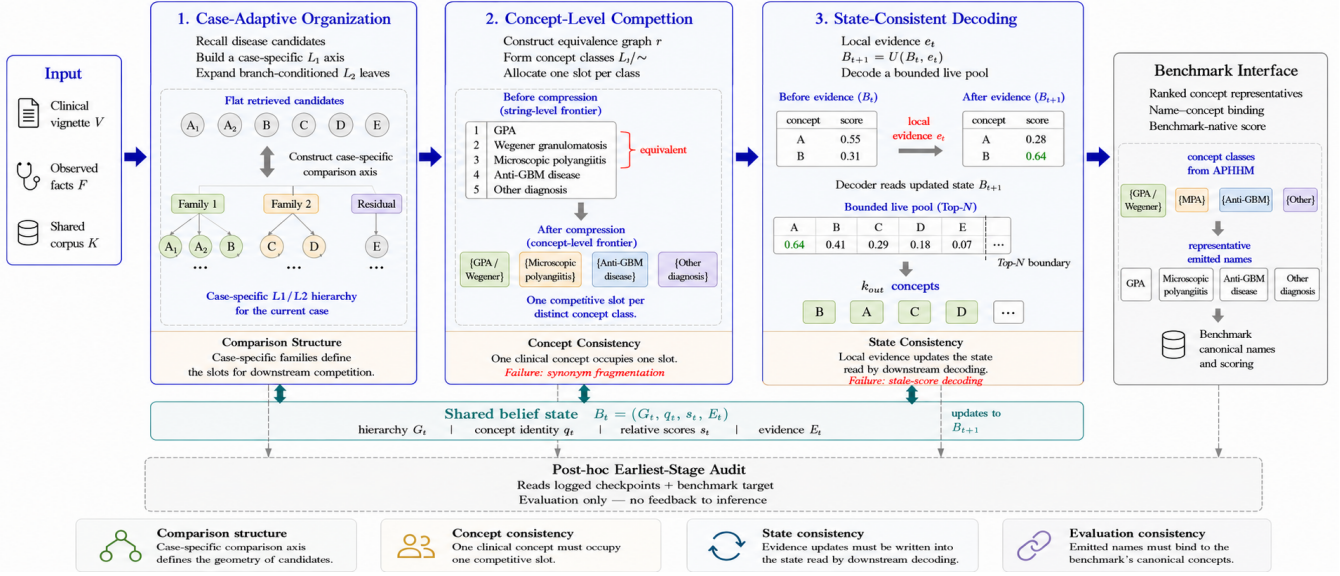


Figure 1: APHHM preserves comparison structure, concept identity, and evidence-conditioned state across stages: recalled diseases are organized on a case-specific L1/L2 axis, equivalence-aware competition allocates frontier slots to clinical concept classes rather than surface strings, and local evidence-conditioned scores are written into the shared belief state before bounded global decoding. The footer bands identify the comparison structure and the requirement enforced at each stage; name–concept binding and earliest-stage attribution are evaluation-only.

4.1 Case-Adaptive Organization and Evidence

APHHM gathers concrete diseases from case-report corpora, guideline and textbook corpora, and zero-temperature LLM enumeration, then combines the ranked lists with reciprocal rank fusion (RRF) (Cormack, Clarke, and Buettcher 2009). An LLM organizes these candidates into L1 families using one case-specific axis, near-exclusive siblings, and a residual family. The partition serves as a temporary comparison coordinate system specialized to the current syndrome. Branch-conditioned retrieval generates L2 disease leaves within each family. When a high-ranked disease has no suitable family, a non-reducing repair step revises the axis to restore structural coverage.

Candidate-relative evidence selection prioritizes facts expected to change the order among live families rather than facts that are merely salient. The implementation uses sparse symmetric updates: each selected fact primarily supports one family and opposes at most one distinct alternative. The selector may abstain when the families do not support a meaningful contrast. The axis comparison therefore evaluates how the representation reshapes the full selector cascade.

4.2 Concept-Level Candidate Competition

Let L be the live leaf set and let $r(\ell_i, \ell_j)$ be the gold-blind pairwise predicate that two labels denote the same diagnosis at the scored granularity. APHHM symmetrizes r and uses connected components of the resulting graph; the induced transitive closure is denoted by $\sim$, and $\pi : L \rightarrow L/\sim$ maps leaves to concept classes. The highest-ranked leaf represents each component. A decision set $S_{k_{out}}$ should allocate its

budget over *classes*, not strings:

$$S_{k_{out}} \subseteq L, \quad |S_{k_{out}}| \leq k_{out}, \quad |\pi(S_{k_{out}})| = |S_{k_{out}}|.$$

When all available distinct slots are filled, $|S_{k_{out}}| = \min\{k_{out}, |L/\sim|\}$. Concretely, if three of the top five leaves belong to one class, the system has spent 60% of its decision budget restating one hypothesis while alternative classes go unexamined.

The resulting frontier contains at most one representative per component and concentrates the output budget on distinct concept classes. Because connected-component closure can join a pair through an intermediate synonym chain, the pairwise predicate, closure rule, and representative selection are separate implementation choices specified in the Supplementary Material. The current routing heuristic chooses between synonym merging and closed-pool calibration to enforce concept-level competition at the appropriate execution site.

Structural property: rank-1 invariance. Under deletion-only compression, retaining the highest-ranked member of every component preserves the global rank-1 leaf. We verified this property over 15,400 perturbed partitions. Section 6.3 therefore evaluates the operator with candidate diversity, open-set coverage, and graded ranking endpoints that respond to compression.

4.3 Shared Belief-State Update and Bounded Decoding

Branch-conditioned L2 generation builds leaves under each live L1 family with per-family budgets and consistency

checks. Local evidence annotation produces a local evidence-conditioned score for each updated leaf, represented abstractly as $B_{t+1} = U(B_t, e_t)$ for selected evidence e_t . Without score write-back, global decoding reads the stale build-time state $D(B_t)$; with **evidence-conditioned score write-back**, it reads $D(B_{t+1})$ after the updated leaf fields have been copied into the shared tree. The updated scores are relative ranking quantities, not calibrated probabilities.

APHHM supports two bounded decoding strategies: a local frontier that retains a second candidate when the within-family margin is small, and an updated-score pool containing the Top- N leaves from the score-updated state. On Open-XDDx, multi-doctor discussion is restricted to the Top- $N=15$ leaves of this pool, and the final $k_{\text{out}}=5$ outputs are projected back into it (Chen et al. 2025b; Yuan et al. 2026). L1 scores enter arbitration as soft priors, while the score-updated leaf state drives final comparison. Independently normalized within-family scores are not treated as globally calibrated probabilities.

4.4 Evaluation Consistency

Benchmarks convert a ranked belief state into a score through an additional name-concept binding stage. On Diagnosis-Arena (DA), the answer mapper can leave an option-to-leaf binding empty even when a synonym-equivalent leaf appears in the ranked shortlist. An offline synonym/bridge repair raises external-baseline Top-1 accuracy by approximately 0.02–0.08 and APHHM by approximately 0.02, demonstrating that binding quality affects all compared systems. We report every system under the original benchmark interface and analyze binding failures separately. MedCaseReasoning (MCR) uses open single-trajectory scoring and directly scores the emitted disease names.

5 Experimental Setup

5.1 Datasets and Evaluation

We use fixed sequential 100-case evaluation subsets from three open-ended diagnostic benchmarks: DiagnosisArena (DA) (Zhu et al. 2026) for option-mapped Top-1/Top-2 accuracy, MedCaseReasoning (MCR) (Wu et al. 2025) for single-trajectory accuracy and reasoning-fact recall, and Open-XDDx (OX) (Zhou et al. 2025) for multi-label micro-F1 at $k_{\text{out}}=5$. Each subset supports a paired mechanism comparison under its benchmark-native metric. For compression, any-hit@5 records whether at least one of the first five delivered diagnoses matches the target; open-MRR@5 is the reciprocal rank of the first match and is zero when no match occurs. Gold labels are never used during generation, evidence selection, state update, or arbitration. All methods use the original benchmark interfaces described in Section 4.4.

5.2 Implementation and Comparisons

All retrieval-based systems use the same indices, and the shared generation backbone is meta-llama/llama-3.3-70b-instruct (Meta AI 2024). MCR reasoning-recall scores use the gemini-2.5-flash endpoint (Google 2025) with a fixed prompt and no lexical substitution. Completed comparison arms include direct CoT (Wei et al. 2022),

CoT+RAG (Lewis et al. 2020; Xiong et al. 2024), flat retrieve-rerank and beam variants, Dual-Inf (Zhou et al. 2025), MDAgents (Kim et al. 2024), MAC (Chen et al. 2025b; Yuan et al. 2026), MEDDxAgent (Rose et al. 2025), self-consistency (Wang et al. 2023), self-refine (Madaan et al. 2023), Medprompt (Nori et al. 2023), MedRAG (Zhao et al. 2025), i-MedRAG (Xiong et al. 2025), and Chain-of-Diagnosis variants (Chen et al. 2025a). All API-based arms use the shared backbone; the DiagnosisGPT-6B local-model row is retained as a separately marked reference point.

To separate structure from inference volume, we use a ten-trajectory flat reranking control with reciprocal rank fusion (Flat rerank $\times 10$ (RRF)), requiring approximately 90 model calls per case (92.4 on DA, 89.8 on OX, and 93.2 on MCR). Each point estimate uses one fixed evaluation pass per configuration; paired resampling estimates uncertainty rather than averaging repeated runs. Candidate enumeration uses temperature zero, while prompts, judge identities, case order, retrieval indices, and scoring interfaces are fixed within each comparison.

5.3 Research Questions and Statistical Tests

The evaluation addresses three questions: **RQ1**: Does a case-adaptive axis improve hypothesis organization? **RQ2**: Does competition over canonical concepts improve a fixed candidate budget? **RQ3**: Does writing local evidence into the shared belief state improve global decoding independently of evidence budget? The reported contrast family contains the axis comparison on DA Top-1, semantic versus count-matched blind compression on MCR any-hit@5, evidence-conditioned score write-back versus no score write-back at fixed budget on OX micro-F1, and a selector negative control. Holm adjustment controls multiplicity across these four contemporaneous contrasts (Holm 1979). We use paired exact tests for binary outcomes and a paired case bootstrap that re-samples cases and recomputes OX micro-F1. Open-MRR@5 is reported descriptively as a paired bounded nonbinary outcome.

Evidence scope. The experiments use fixed 100-case subsets for mechanism analysis. RQ1 estimates the effect of changing the case axis within the deployed selector cascade; RQ2 tests partition semantics on audited open-set MCR endpoints, with open-MRR@5 descriptive; and RQ3 isolates score write-back within the fixed OX decoder and evidence-budget design. The leaf-injection arm is a retrospective diagnostic from an earlier judge configuration. The flat control matches model-call counts; token, latency, and monetary cost remain future audit dimensions.

6 Results

6.1 Overall Performance

APHHM ranks first in all six columns among the completed shared-backbone arms in Table 2. On the benchmark-native end tasks, it exceeds the strongest external result by +0.09/ + 0.07 on DA Top-1/Top-2, +0.26 on MCR accuracy, and +0.081 on OX micro-F1. Its MCR reasoning recall is also highest (0.753 vs. 0.570), consistent with a belief state

Method	DA (Zhu et al. 2026)			MCR (Wu et al. 2025)		OX (Zhou et al. 2025)
	@1	@2	MRR	Acc	R-Recall	micro-F1
APHHM	0.71	0.78	0.748	0.50	0.753	0.651
MEDDxAgent	0.62	0.71	0.665	0.24	0.412	0.491
MAC	0.61	0.67	0.640	0.23	0.527	0.570
Dual-Inf	0.60	0.70	0.650	0.17	0.444	0.507
i-MedRAG	0.60	0.67	0.635	0.19	0.482	0.419
MDAgents	0.58	0.67	0.625	0.20	0.570	0.543
Self-refine	0.57	0.62	0.595	0.21	0.447	0.530
Flat rerank	0.56	0.63	0.595	0.17	0.404	0.495
CoT+RAG	0.55	0.63	0.590	0.22	0.478	0.466
Direct CoT	0.54	0.61	0.575	0.18	0.510	0.543
Chain-of-Diagnosis + shared corpus	0.54	0.55	0.545	0.14	0.430	0.475
Self-consistent CoT (5 samples)	0.52	0.61	0.565	0.19	0.369	0.539
Flat beam search	0.52	0.60	0.560	0.21	0.447	0.510
Medprompt-style	0.52	0.57	0.545	0.18	0.294	0.522
MedRAG	0.48	0.52	0.500	0.17	0.557	0.485
Flat rerank $\times 10$ (RRF)	0.47	0.59	0.530	0.15	0.383	0.487

Table 2: Completed shared-backbone arms on the fixed 100-case subsets under unmodified benchmark interfaces. DA MRR is MRR@2; R-Recall is LLM reasoning recall; bold marks the best value per column. The separately backboned DiagnosisGPT-6B row is in the Supplementary Material.

that preserves diagnostically relevant facts. The mechanism tests below attribute these margins to organization, concept competition, and state propagation.

6.2 RQ1: Case-Adaptive Organization

We compare the case-adaptive L1-axis pipeline with fixed ICD/specialty and case-irrelevant random axes while preserving family count and candidate budget. As shown in Table 3, changing the axis within the current pipeline raises DA Top-1 accuracy by 0.34 relative to the random axis (paired discordance 40/6, exact $p = 3.1 \times 10^{-7}$, Holm-adjusted $p = 1.2 \times 10^{-6}$ across the four contemporaneous contrasts) and by 0.20 relative to the fixed taxonomy (25/5, $p = 3.2 \times 10^{-4}$). The fixed taxonomy emits more leaves than the case-adaptive system (30.1 vs. 17.8 per case), showing that the case-adaptive gain accompanies more effective organization of a smaller frontier.

The same candidate-relative evidence selector runs under every axis. Mean selected L1 facts decrease from 4.94 under the case-adaptive axis to 0.65 under the fixed taxonomy and 0.13 under the random partition; empty rankings increase from 1 to 11 and 45, respectively. Thus, the intervention changes both grouping and whether candidate-relative evidence can distinguish the resulting families. Conditioning on nonempty rankings largely removes the random-axis difference (7/6) but leaves a fixed-taxonomy difference (17/5, $p = 0.017$), suggesting both a selector-mediated effect and a residual organizational effect.

Replacing the evidence selector with a salience-based alternative, or removing its discrimination hints, changes Top-1 accuracy by only 0.04 (both $p > 0.4$). Together with the axis intervention, these comparisons locate the large gain in the compatibility between case-adaptive organization and downstream candidate-relative selection.

6.3 RQ2: Concept-Level Candidate Competition

Because concept compression retains the best-ranked member of each equivalence class, its rank-1 output is invariant to class membership (Section 4.2). We verify endpoint sensitivity with 200 randomized re-partitions per case: Top-1 is an algebraic identity on both MCR and DA, whereas MCR any-hit@5 and open-MRR@5 respond to changes in equivalence structure. The analysis therefore uses these open-set endpoints together with aggregate accuracy over the 98 commonly scored MCR cases.

Equivalence handling occurs during tree construction and again during decision-time compression. Table 4 crosses the two sites.

Removing equivalence handling from both sites lowers aggregate accuracy from 0.50 to 0.42; the paired difference is +10.2 percentage points with exact interval [+3.9, +10.2] over 10/0 discordant pairs (unadjusted $p = 0.002$). Removing either site alone leaves accuracy at 0.50, indicating that build-time and decision-time handling are partially substitutable rather than independently additive.

Decision-time compression removes 56.5% of surviving candidates (4.64 $\rightarrow$ 2.02 per case) while bounding the paired binary accuracy difference to $[-1.0, +1.0]$ percentage points. Open-MRR@5 increases descriptively from 0.52 to 0.53. A count-matched blind merge is worse on both responsive endpoints (any-hit@5 0.475 ± 0.039 vs. 0.571 ; open-MRR@5 0.444 ± 0.014 vs. 0.486), where $\pm$ denotes the standard deviation over 200 randomized partitions. The partition-randomization result ($p = 0.015$; Holm-adjusted $p = 0.030$) shows that semantic class membership matters beyond the number and sizes of compressed groups. The blind-merge null is nondegenerate: the synonym graph is complete in 0/35 perturbable DA cases and 0/42 perturbable MCR cases, 82.8%/85.1% of draws violate the quotient, and each draw averages 1.99 false merges and 1.98 false splits; the full audit

Axis	Evaluation set	@1	@2	Δ @1	disc. b/c
Case-adaptive (full model)	full	0.71	0.78	—	—
Fixed ICD/specialty	full	0.51	0.60	+0.20	25/5
Random (case-irrelevant)	full	0.37	0.42	+0.34	40/6
Fixed ICD/specialty	nonempty ($n=89$)	0.57	—	+0.135	17/5
Random (case-irrelevant)	nonempty ($n=55$)	0.67	—	+0.018	7/6

Table 3: L1-axis comparisons on DA ($n=100$). Δ is the case-adaptive minus the comparison score; nonempty rows condition on cases that produced a ranking.

Configuration	Sites	Acc@1	any-hit@5	open-MRR	cand./case
Full model	both on	0.50	0.58	0.53	2.02
No decision-time routing	build only	0.50	0.58	0.52	4.64
No build-time de-duplication	decision only	0.50	0.55	0.52	1.81
Neither	none	0.42	0.56	0.48	2.11

Table 4: Equivalence handling on MCR with build-time and decision-time sites crossed ($n=98$).

is in the Supplementary Material. Blind merge leaves 0.79 synonymous duplicate pairs, whereas the deployed operator’s 0.00 is guaranteed by construction.

DA provides a complementary binding audit. Closed-set rematching reports a 0.112 Top-1 increase even though the two configurations emit identical rank-1 labels in all 89 gated cases and the open-set Top-1 difference is exactly zero. The contrast localizes this increase to member-to-representative option binding, while the responsive open-set MCR end-points measure changes in concept competition.

Figure 2 separates the two questions that the compression ablations answer. Substituting one policy or component for another stays inside the pre-registered ± 5 percentage-point non-inferiority margin in every case, whereas removing equivalence handling altogether clears it. The narrow bounds follow from the paired design rather than from limited power, because a paired difference cannot exceed the discordant-pair count divided by n ; reporting them as bounds therefore states how small an effect can be instead of failing to detect one. On DA, the stricter concept-identity predicate scores 0.57 versus 0.71 and produces more clusters in 37 cases and fewer in none; the closed-set member-to-representative audit explains this cross-caliber difference. Across these analyses, the stable mechanism is competition over concept classes, with several routing realizations reaching similar binary accuracy.

6.4 RQ3: State Consistency and Bounded Decoding

Table 5 crosses evidence budget B_{evid} with evidence-conditioned score write-back while holding the per-family cap and global decoder fixed.

At the fixed budget, evidence-conditioned score write-back increases micro-F1 by 0.075, with a paired bootstrap 95% interval of $[0.036, 0.113]$ and $p = 4.0 \times 10^{-4}$ (5000 resamples; Holm-adjusted $p = 1.2 \times 10^{-3}$). A per-case sign test agrees in direction (42/14, $p = 2.3 \times 10^{-4}$). By contrast, increasing the evidence budget changes F1 by at most 0.008 with or without score write-back, and the interaction

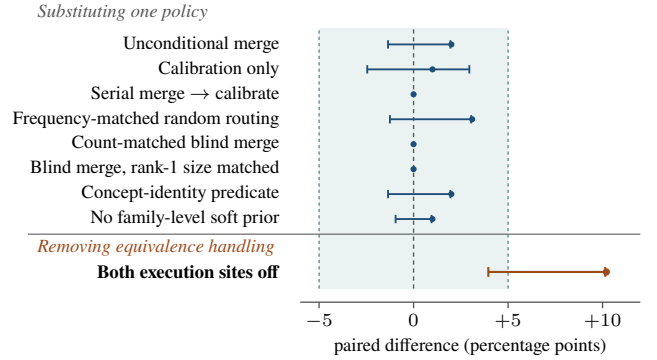


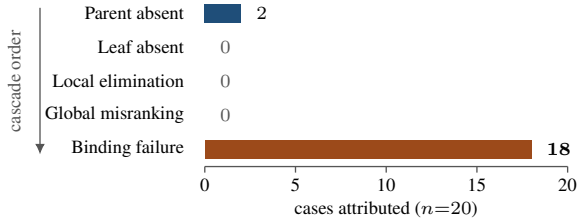
Figure 2: Paired bounds for the equivalence-compression variants on MedCaseReasoning accuracy ($n=98$, full model minus variant). Shading marks the pre-registered ± 5 percentage-point non-inferiority margin. Intervals are exact conditional intervals over discordant pairs, unadjusted for multiplicity; a point estimate on the upper bound means every discordant pair favors the full model, and the three variants without an interval have no discordant pair.

is -0.001 , so within this OX configuration the write-back accounts for the main gain.

Holding the score-updated tree fixed, closed-panel decoding reaches micro-F1 0.651, compared with 0.610 for the updated-score-pool decoder and 0.593 for direct updated-score Top-5. Paired case-F1 favors the closed panel over direct updated-score Top-5 by +0.056 (bootstrap 95% CI $[0.028, 0.085]$) and over the updated-score-pool decoder by +0.040 (CI $[0.013, 0.069]$). Forcing one candidate per family (0.657) and removing the family cap (0.649) both remain close to the full model (0.651), making the cap empirically secondary in this configuration. An emit-enabled variant scores similarly to score write-back alone (0.645 vs. 0.651), further localizing the gain to belief-state updates.

Evidence budget	No write-back	Score write-back
Fixed ($B_{\text{evid}}=4$)	0.576	0.651
Default ($B_{\text{evid}}=6$)	0.584	0.658

Table 5: OX micro-F1 under a 2×2 comparison of evidence budget and evidence-conditioned score write-back ($n=100$).



Counterfactual. Acting on the unstratified reading and injecting leaves across the full tree adds 16.1 leaves per case and raises coverage to 0.93, yet Top-1 accuracy falls **0.72** $\rightarrow$ **0.42** (39/9 discordant pairs, $p=1.5 \times 10^{-5}$).

Figure 3: Stage attribution of the 20 apparent coverage misses on DiagnosisArena; the evaluation-interface stage is in the second colour. Under earliest-stage assignment the three intermediate stages receive no cases, and reading the same misses as insufficient recall motivates the intervention below, which enlarges the candidate space and lowers accuracy. The audit is retrospective.

6.5 Evaluation Analysis: Stage Attribution

A retrospective stage audit of the 20 apparent coverage misses on DA attributes 18 to binding failure and 2 to absence of a suitable L1 family, so nine tenths of an apparent recall gap is an evaluation-interface failure (Figure 3). Treating all 20 as recall failures would suggest generating more leaves, yet full-tree leaf injection lowers Top-1 accuracy from 0.72 to 0.42, which is why stage attribution should precede a recall intervention.

6.6 Ten-Trajectory Flat Reranking Control

Flat retrieve–rerank performance does not improve with inference volume. Moving from its native configuration (about two model calls per case) to a structural proxy (about nine) and to Flat rerank $\times 10$ (RRF) (about 90) gives 0.56/0.48/0.47 Top-1 on DA, 0.495/0.479/0.487 micro-F1 on OX, and 0.17/0.17/0.15 accuracy on MCR, whereas APHHM reaches 0.71, 0.651, and 0.50. The Supplementary call audit shows that the ten-trajectory control is matched within 2% on DA and is call-richer on OX and MCR. The persistent margin at similar or higher call counts indicates that structured hypothesis management contributes beyond repeated flat sampling.

7 Discussion

The results establish a state-management view of hierarchical reasoning that complements fixed specialty routing. The case-specific hierarchy provides a coordinate system

for comparing hypotheses, concept-level competition prevents surface forms from exhausting the decision budget, and evidence-conditioned score write-back keeps locally used evidence available globally. These three principles sit above prompt-level implementation choices, and they motivate extensions to other open-ended tasks with synonymous hypotheses and multi-stage local/global decisions.

The evaluation analysis also has implications beyond APHHM. When an offline name-matching pass moves accuracy for several systems, the reported score reflects both hypothesis quality and normalization coverage. Benchmarks should therefore specify a common name–concept binding contract, and papers should report mapping failures separately from generation and ranking failures.

The transferable contribution is therefore a coherent representation of concept identity and belief state throughout the reasoning trajectory, realized here by case-dependent organization compatible with downstream selection, equivalence-aware competition, and evidence-conditioned score write-back.

Future validation should extend the mechanism tests to independently sampled and full-benchmark evaluations, cross axis and selector choices factorially, replicate evidence-conditioned score write-back across datasets, and validate the reasoning-recall judge against independent human or model judgments. Token-, latency-, and cost-level resource audits would complement the current call-count analysis, while larger samples would resolve smaller design choices and nonbinary endpoints.

8 Conclusion

Open-ended differential diagnosis requires more than a sufficiently broad candidate list: candidates must compete as clinical concepts, local evidence must update the state used by global decoding, and evaluation failures must be distinguished from reasoning failures. APHHM implements these requirements through case-adaptive organization, equivalence-aware competition, evidence-conditioned score write-back, and bounded decoding. Across three diagnostic benchmarks, the comparisons support an axis-conditioned pipeline effect, concept-aware competition on audited open-set endpoints, and score write-back on Open-XDDx; APHHM retains its end-task margins against a similar-call-count flat pipeline. The broader implication is that consistent hypothesis identity and belief state convert recall into reliable downstream decisions.

Ethical Statement

This study uses public research vignettes and does not involve patient care or clinical deployment. The evaluated models may reproduce demographic, geographic, and documentation biases present in their training data and retrieval corpora. The LLM-based reasoning-recall judge may add model-specific bias; its identity and prompt are fixed across comparisons, but independent human validation remains necessary. The system should not be used for diagnosis without prospective clinical evaluation, calibrated uncertainty, privacy safeguards, and expert oversight.

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